

Lab 1 Report: Pediatric Traumatic Brain Injury Prediction Models and Data Exploration

Introduction

Every day, numerous pediatric patients present to emergency departments (EDs) with head trauma. While the majority of these cases are mild and do not necessitate advanced imaging, a small fraction involve clinically important traumatic brain injuries (ciTBI). Undiagnosed ciTBI can lead to severe complications, making accurate triage essential. A primary clinical challenge is identifying low-risk patients who can safely forgo computed tomography (CT) scans. Although CT scans accurately detect intracranial injuries, they expose children to ionizing radiation. The risk of lethal malignancies from pediatric cranial CT scans is estimated between 1 in 1,000 and 1 in 5,000, with the risk increasing for younger children (Kuppermann et al., 2009). Therefore, clinical decision-making must carefully balance the benefits of imaging against these risks.

This report explores the Pediatric Emergency Care Applied Research Network (PECARN) dataset, which contains clinical information and outcomes for children with head trauma. The study performs comprehensive exploratory data analysis (EDA) to understand the relationships between clinical presentation, CT utilization, and ciTBI risk. Furthermore, I develop and evaluate predictive models, including the PECARN clinical decision rule, logistic regression, and decision trees, to identify children at low risk of ciTBI.

Part I: Data Exploration and Visualization

Data Overview

The dataset originates from a study aimed at developing and validating the PECARN clinical decision rule for predicting ciTBI in children. Patients were enrolled across 25 emergency departments. Enrollment occurred in two phases: a derivation cohort (June 2004 to March 2006) and a validation cohort (March to September 2006). The working dataset, after initial filtering, contains 42,412 eligible patients who presented within 24 hours of injury with Glasgow Coma Scale (GCS) scores of 14-15. The cohort is stratified by age: approximately 25% (10,718) are under 2 years old, and 75% (31,694) are 2 years or older. The primary outcome, ciTBI, is a rare event occurring in only 0.89% (376 out of 42,412) of the cohort, while CT imaging was performed in 35.3% (14,969) of patients.

To prepare the raw data for analysis and prevent data leakage, an extensive data cleaning pipeline was implemented. There are some data quality challenges, including coded values, high rates of missingness, outliers, complex parent-child dependencies, and internal inconsistencies. Some key data cleaning and preprocessing steps I did are:

- Standardizing column names to a consistent format for easier handling.
- Dataset-specific codes indicating “Not Applicable” (92) were converted to missing values, as they do not provide predictive information for ciTBI. Codes for “Other” (90) and “Pre-verbal/Non-verbal” (91) were preserved as valid, distinct categories.
- Parent-Child Dependency inconsistencies were resolved. For example, if a parent variable was recorded as “No,” any conflicting detailed child variables were set to missing to prevent inaccurate risk inflation.
- CT-dependent fields (e.g., CT findings, Indicator variables) and many others (e.g. demographics) were cleared for patients who did not undergo a CT scan to avoid target leakage into the predictor space.
- Recalculations were performed for GCS Total to ensure the recorded GCS total matched the sum of the eye, verbal, and motor components, flagging inconsistencies for auditing.
- The ciTBI outcome was reconstructed from component variables to ensure consistency with the original criteria, and discrepancies were flagged for review.
- For patients without CT scans, all CT-dependent fields were set to missing to prevent leakage and maintain data integrity.

The dataset has several limitations that affect both analysis and interpretation. First, many variables contain substantial missing data, including clinically important predictors. Second, the primary outcome, clinically important traumatic brain injury (ciTBI), must be reconstructed from multiple fields, and minor inconsistencies in the raw variables introduce some uncertainty in the exact target definition. Third, ciTBI is rare, occurring in less than 1 percent of patients, which creates class imbalance and increases the risk of unstable estimates or overfitting despite attempts to mitigate this through modeling choices.

Next, I perform exploratory data analysis to understand the relationships between clinical features, CT use, and ciTBI risk. I examine the distribution of key variables, identify potential predictors of ciTBI, and explore how CT use varies across different patient profiles.

Finding 1: Symptom Profiles, CT Use, and the Significance of Missingness

The PECARN rule uses a small number of binary clinical predictors to triage children with minor head trauma. Rather than analyzing predictors individually, I treated the six PECARN predictors as a joint “symptom profile” for each patient within each age stratum. This allows three clinically relevant questions: How does risk accumulate across predictors? How does CT use vary across similar-risk profiles? Does missingness in key predictors behave like random noise or like a clinically meaningful subgroup?

To align with the age-stratified design of the PECARN rules, all analyses were separated into cohorts of children under 2 years and those 2 years and older. Tables 1 and 2 show CT use and observed ciTBI rates by number of positive PECARN predictors.

Across both age groups: (1) CT use increases monotonically as predictors accumulate. (2) Observed ciTBI risk increases sharply as predictors accumulate.

Table 1: Table 1. CT use and observed ciTBI rates by number of PECARN predictors present (age < 2)

Number of positive predictors	Number of patients	CT rate (%)	ciTBI rate (%)
0	5728	14.80	0.02
1	3254	36.70	0.83
2	1299	69.70	3.46
3+	620	86.80	12.90

Table 2: Table 2. CT use and observed ciTBI rates by number of PECARN predictors present (age ≥ 2)

Number of positive predictors	Number of patients	CT rate (%)	ciTBI rate (%)
0	18150	13.20	0.05
1	9273	58.60	1.15
2	3804	88.40	5.34
3+	1251	95.90	23.26

- Age < 2 (Table 1): CT use increased from 14.80% (0 predictors) to 86.80% (3+ predictors). Observed ciTBI increased from 0.02% to 12.90%.
- Age ≥ 2 (Table 2) CT use increased from 13.20% (0 predictors) to 95.90% (3+ predictors). Observed ciTBI increased from 0.05% to 23.26%.

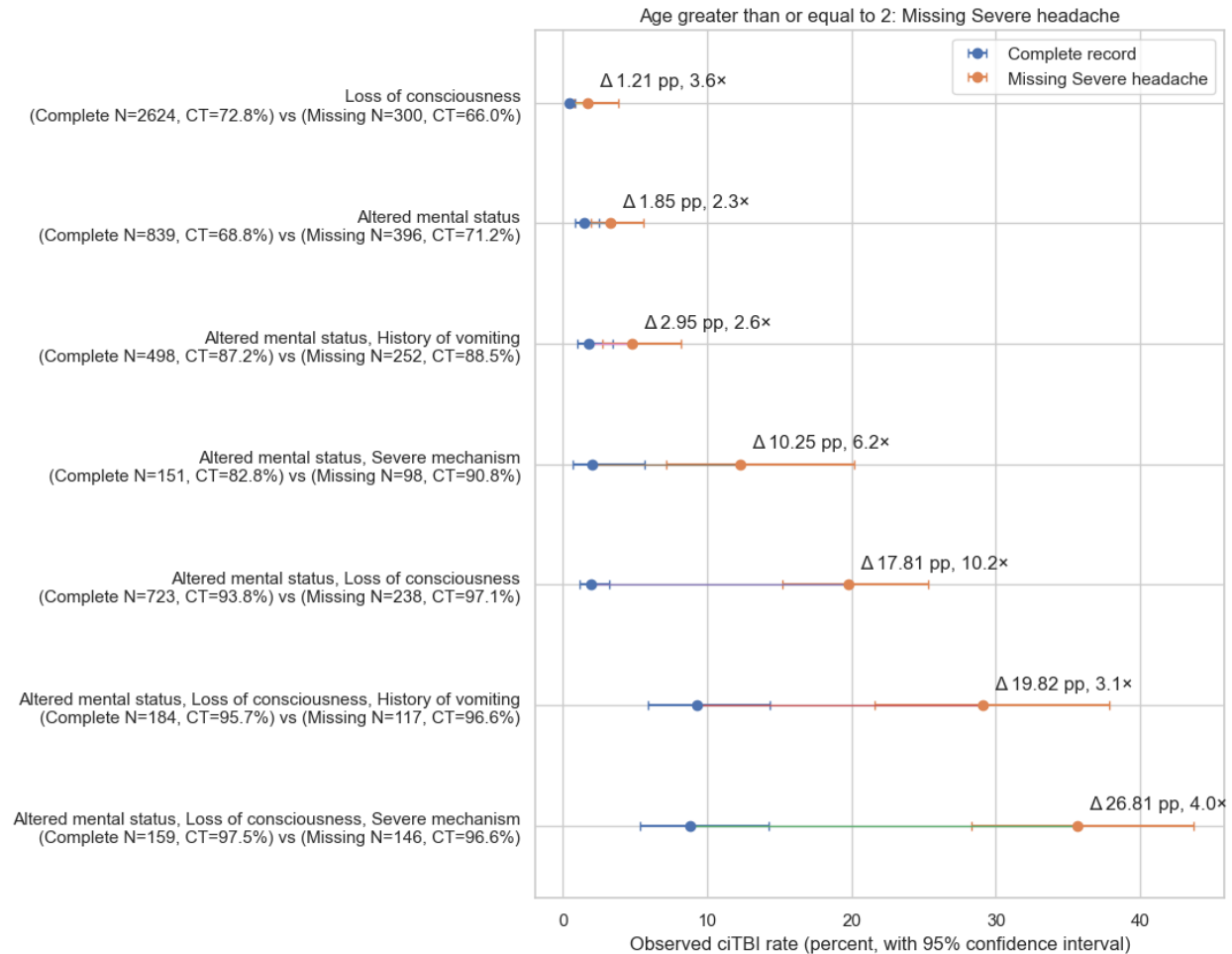
First, this confirms internal consistency. The PECARN predictors behave directionally as intended. Risk is very low when no predictors are present and increases dramatically as predictors accumulate. Second, the slope of risk increase is steeper in children age 2 or greater. This suggests that the combination of predictors in older children separates high-risk from low-risk patients more strongly than in infants and toddlers, where developmental limitations affect symptom assessment. Crucially, the most significant insight from this symptom profiling relates to how missing clinical fields are handled.

Within the same observed predictor set, I compared:

- Complete record: predictor values observed (no “missing” flag for that symptom).
- Missing [symptom]: same predictor set, but that symptom field is missing and is flagged as missing.

The plots show observed ciTBI rates with Wilson 95% confidence intervals. Each comparison holds all other predictors fixed. If missingness were random measurement noise, ciTBI rates would be similar between complete and missing subgroups. However, that was not the case.

Missing severe headache in age 2 and above



- Δ = absolute difference in ciTBI in percentage points (pp) between missing and complete.
- $\times$ = ratio of ciTBI rates (missing divided by complete).

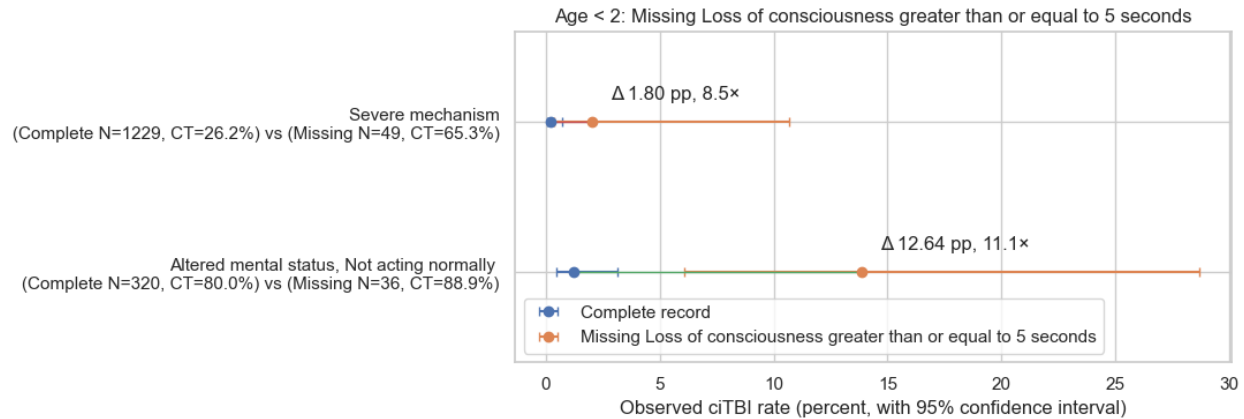
For profiles involving altered mental status and loss of consciousness, the increases become large:

- Altered mental status + loss of consciousness: Δ 17.81 pp, 10.2×
- Altered mental status + loss of consciousness + vomiting: Δ 19.82 pp, 3.1×
- Altered mental status + loss of consciousness + severe mechanism: Δ 26.81 pp, 4.0×

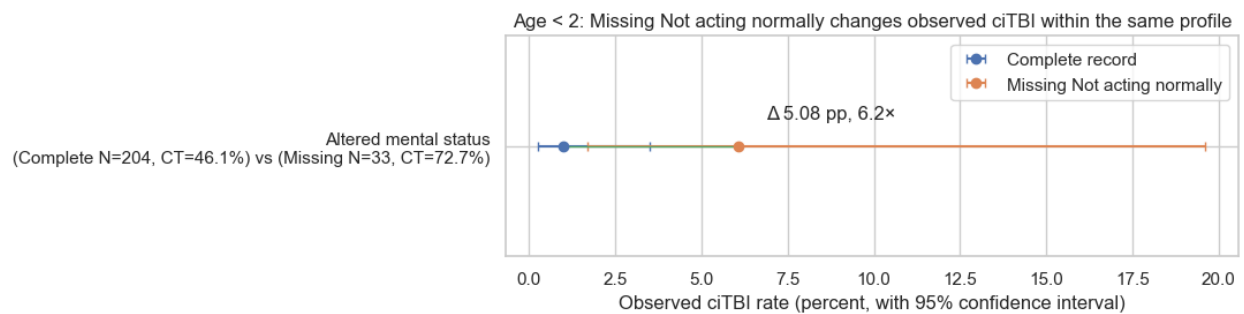
These are large shifts within already high-risk strata. Even in lower-risk profiles (e.g., LOC only), missing severe headache is associated with a positive risk shift, though smaller. This pattern indicates that “missing severe headache” is not neutral. It marks a subgroup with higher observed ciTBI, even after conditioning on the same visible predictors.

Missing LOC duration and Not acting normally in age < 2

In younger children, missing LOC duration and not acting normally reported by parents show a similar pattern.



Among toddlers with severe mechanism, missing LOC duration increases ciTBI by 1.80 percentage points (8.5× relative increase). In more concerning profiles such as altered mental status + not acting normally, the increase is much larger (Δ 12.64 percentage points).



When altered mental status is present, missing “not acting normally” increases ciTBI by 5.08 percentage points (6.2×). This missingness likely reflects situations where caregiver report is unavailable or unreliable (no parent present, unclear baseline, limited history). In infants and toddlers, caregiver-assessed variables are central to PECARN. When they are missing, assuming “normal behavior” is risky if the goal is sensitivity.

Across both age groups, missingness in key symptom predictors behaves like a structured subgroup, not random noise. Within the same observed PECARN predictor set, patients with missing symptom fields consistently show higher observed ciTBI rates than complete records. In other words, missingness carries information. Some implications of this finding are:

- Treating unknown as low risk can systematically reduce sensitivity.
- Dropping incomplete records can disproportionately remove higher-risk cases.
- A conservative rule should treat certain missingness patterns as not low risk, or explicitly encode “unknown” as its own state.

Finding 2: CT Use and Indicators Variables

The clinical challenge extends beyond predicting ciTBI to understanding the behavioral drivers behind a clinician’s decision to order a CT scan. Imaging decisions may change during the ED encounter as new information emerges. Therefore, understanding these decision switches helps to check whether PECARN-style predictors align with actual clinician behavior and if there’s potential selection effects if CT-related variables are later used in modeling.

I used two variables: CTForm1 and CTDone. Then, I grouped each patient into one of four decision states: (1) planned and done (plan=1, done=1). (2) planned not done (plan=1, done=0). (3) not planned not done (plan=0, done=0). (4) not planned but done (plan=0, done=1).

Table 3: Table 3. Cross-tabulation of CTForm1 and CTDone.

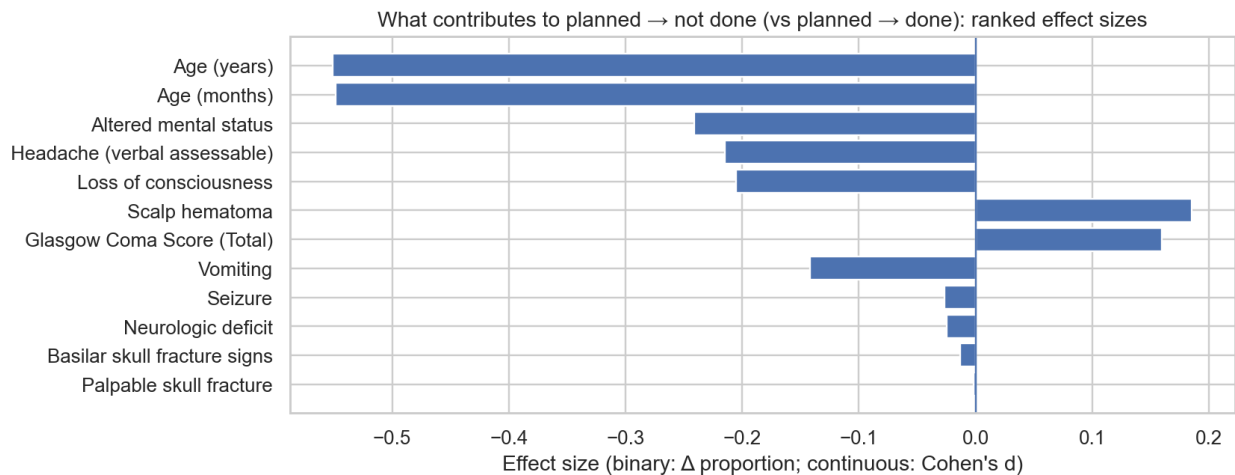
	CT not done	CT done
CTForm1 = 0	26,260	184
CTForm1 = 1	1,210	15,697
CTForm1 = NA	26	2

From this, I derived conditional probabilities that summarize the relationship:

- $P(\text{CT done} \mid \text{CTForm1} = 1) = 92.8\%$
- $P(\text{CTForm1} = 1 \mid \text{CT done}) = 98.8\%$

While CTForm1 largely functions as a proxy for an initial plan to image, the data reveals a meaningful de-escalation subgroup (planned then not done) and a smaller escalation subgroup (not planned then done). More specifically, the observed ciTBI rates by the two planned groups are (1) Planned and done: 0.047334 (about 4.73%) (2) Planned not done: 0.006612 (about 0.66%). This is a large absolute gap (about 4.07 percentage points). It suggests that when clinicians decide not to proceed with a planned CT, they are usually doing so in a subgroup that looks less concerning after further evaluation.

To understand why, I examined covariate differences.



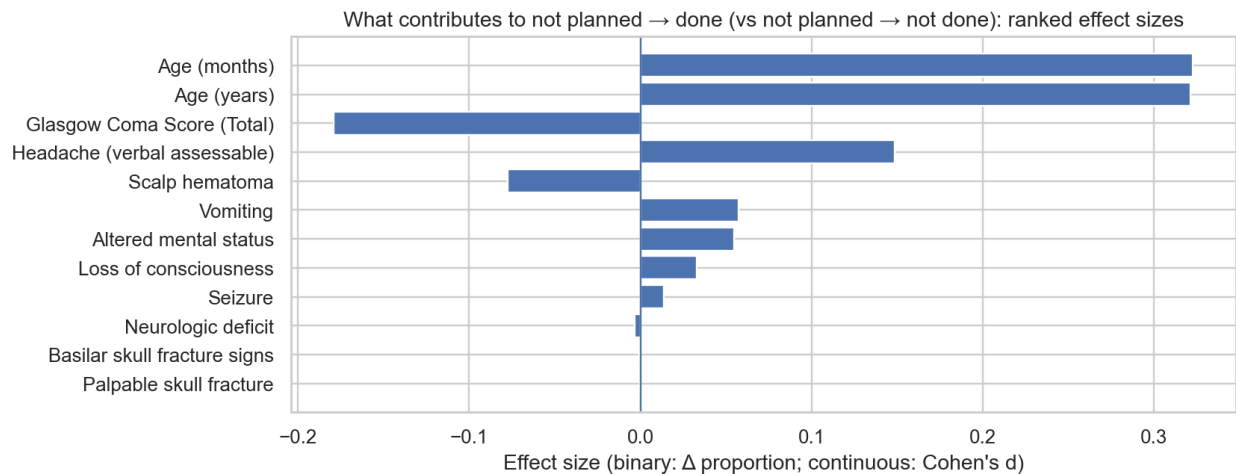
The planned-not-done group differs from planned-and-done primarily in:

- Age (years and months) has the largest negative effect size. The planned-not-done group is older.
- Altered mental status (ams) is strongly negative. The de-escalated group has lower AMS.
- Headache and LOC are negative. Lower prevalence in planned-not-done.
- Vomiting is also negative.
- GCS total is positive. Planned-not-done has higher GCS.
- Hematoma is slightly positive. That suggests hematoma is more common in planned-not-done than planned-and-done, which is interesting and worth noting.

- Skull fracture signs and neuro deficit are near zero.

These shifts demonstrate a coherent clinical pattern primarily driven by neurological stability. The separation between the groups is clearly defined by older age, higher Glasgow Coma Scale (GCS) scores, and lower incidences of both altered mental status (AMS) and loss of consciousness (LOC). Ultimately, this indicates that clinicians actively de-escalate their imaging plans when a patient's objective neurological risk markers prove to be weaker than initially suspected.

The escalation group (not planned then done) is much smaller, but informative.



Compared with not-planned-and-not-done patients, the escalated group shows:

- Age has the largest positive effect size. Escalated patients are older.
- Headache is strongly positive.
- Lower GCS (negative effect size for GCS total) separates the escalated group.
- Hematoma is positive.
- Vomiting and AMS are modestly positive.
- Skull fracture and neuro deficit are minimal.

These are clinically plausible triggers for imaging. Therefore, escalations are not random. They likely reflect new information obtained after initial documentation, or worsening symptoms during observation.

Together, these results show that imaging decisions are dynamic and responsive to clinical signals. CTForm1 behaves like an early intention to image. CTDone reflects final judgment after additional information or reassessment. Furthermore, the de-escalation subgroup has substantially lower ciTBI than the planned-and-done group. Thus, this finding has two implications:

- CTForm1 and CTDone are not interchangeable.
- CT-related variables embed clinician judgment and should not be treated as independent predictors of ciTBI without caution. They reflect a complex interplay of initial assessment, evolving clinical information, and decision-making processes. Using them in modeling requires careful consideration of timing and potential confounding.

Finding 3: Preverbal/Nonverbal

Some PECARN predictors rely on patient report, particularly headache. In the raw dataset, code 91 indicates “Preverbal/Nonverbal,” meaning the child cannot verbalize symptoms. While common in pediatric populations, this code creates clinical ambiguity. The primary analytical question is whether ‘91’ represents standard missing data or distinctly flags a meaningful clinical sub-population. If it carries information about severity or workflow, then treating it as symptom absence or dropping those records could bias risk estimation and reduce sensitivity.

In the full cohort, 91 is associated with substantially higher ciTBI rates.

Table 4: Table 4. ciTBI rates by headache category and age group.

Age group	Headache category	Number of patients	Number of ciTBI events	ciTBI rate (%)
Two Plus	No	14875	66	0.44
Two Plus	Preverbal/Nonverbal	4266	260	6.09
Two Plus	Yes	12773	177	1.39
Under 2	No	1002	3	0.30
Under 2	Preverbal/Nonverbal	9787	142	1.45
Under 2	Yes	25	0	0.00

However, the association mostly disappears in an objective very-low-risk subset. When I restricted to an “objective very-low-risk” subgroup, ciTBI rates become tiny and the groups look much closer.

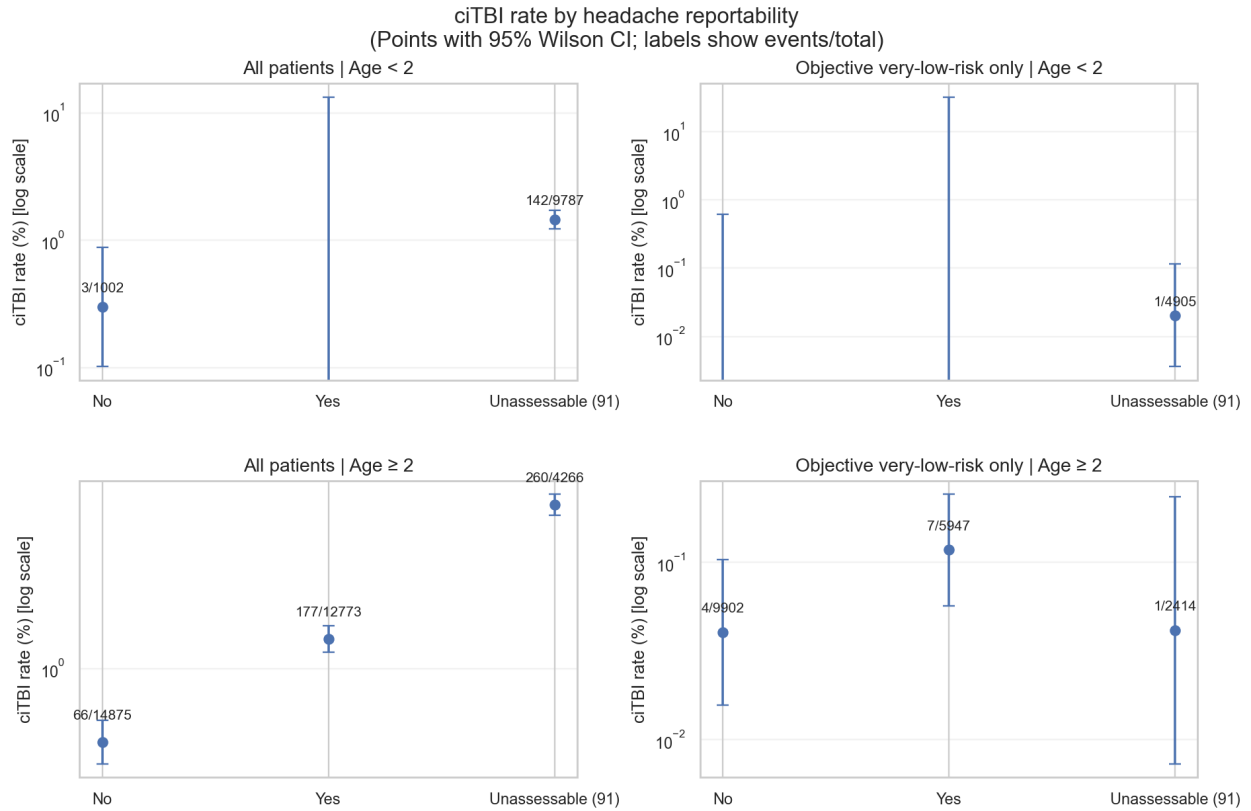
Note: I defined an objective very-low-risk subgroup using the PECARN “very low risk” logic, applied separately by age group. First, I constructed binary indicators that match the clinical predictors used in the PECARN rules (for example, altered mental status from GCS < 15 or AMS = 1, severe mechanism from injury severity, and loss of consciousness 5 or more seconds only when LOC was present or suspected). Then I labeled a child as objective very-low-risk if they had none of the PECARN risk predictors for their age stratum. For children age < 2, this required no altered mental status, no palpable skull fracture, no non-frontal scalp hematoma, no LOC 5 or more seconds, no severe mechanism, and the child acting normally per the parent. For children age 2 or more, this required no altered mental status, no signs of basilar skull fracture, no loss of consciousness, no history of vomiting, and no severe mechanism. To avoid misclassifying incomplete records as low risk, I treated missing values for any of these predictors as not low risk by setting unknown predictor values to “True” (risk-present) when constructing the low-risk flag.

Table 5: Table 5. ciTBI rates by headache reportability and age group in the full sample.

Age group	Headache	n	Events	ciTBI rate (%)
Age ≥ 2	No	14,875	66	0.440
Age ≥ 2	Preverbal/Nonverbal	4,266	260	6.090
Age ≥ 2	Yes	12,773	177	1.380
Age < 2	No	1,002	3	0.300
Age < 2	Preverbal/Nonverbal	9,787	142	1.450
Age < 2	Yes	25	0	0.000

Table 6: Table 6. ciTBI rates by headache reportability and age group in the objective very-low-risk subgroup.

Age group	Headache	n	Events	ciTBI rate (%)
Age ≥ 2	No	9,902	4	0.040
Age ≥ 2	Preverbal/Nonverbal	2,414	1	0.041
Age ≥ 2	Yes	5,947	7	0.118
Age < 2	No	620	0	0.000
Age < 2	Preverbal/Nonverbal	4,905	1	0.020
Age < 2	Yes	8	0	0.000



0/25

In the broader dataset, the “91” (Preverbal/Nonverbal) code consistently correlates with a substantially higher risk of injury. However, once we control for objective low-risk criteria, this excess risk almost entirely disappears. This tells us something important: the “91” code is not an independent physiological predictor of injury, but rather a marker of the clinical complexity and documentation context of the encounter. If our primary goal is sensitivity, the practical takeaway is clear: we cannot safely treat unassessable symptoms as negative findings. Because these unassessable scenarios frequently occur in higher-acuity encounters, defaulting them to “no symptom” would artificially lower the risk profile of a subgroup that actually contains objective danger signs. Conversely, the “91” code alone shouldn’t automatically flag a child as high-risk if

their objective clinical markers are otherwise completely reassuring. While it prevents us from ruling out injury based on symptoms alone, it doesn't automatically mandate an escalation in care.

Ultimately, while the 'Preverbal/Nonverbal' label acts as an informative severity marker in the full cohort, its apparent predictive power fades when objective factors are taken into account. Therefore, a conservative and safe clinical rule should treat unassessable symptoms as a state of clinical uncertainty that prevents a "low-risk" classification, rather than mistaking an absence of evidence for evidence of absence

Stability check

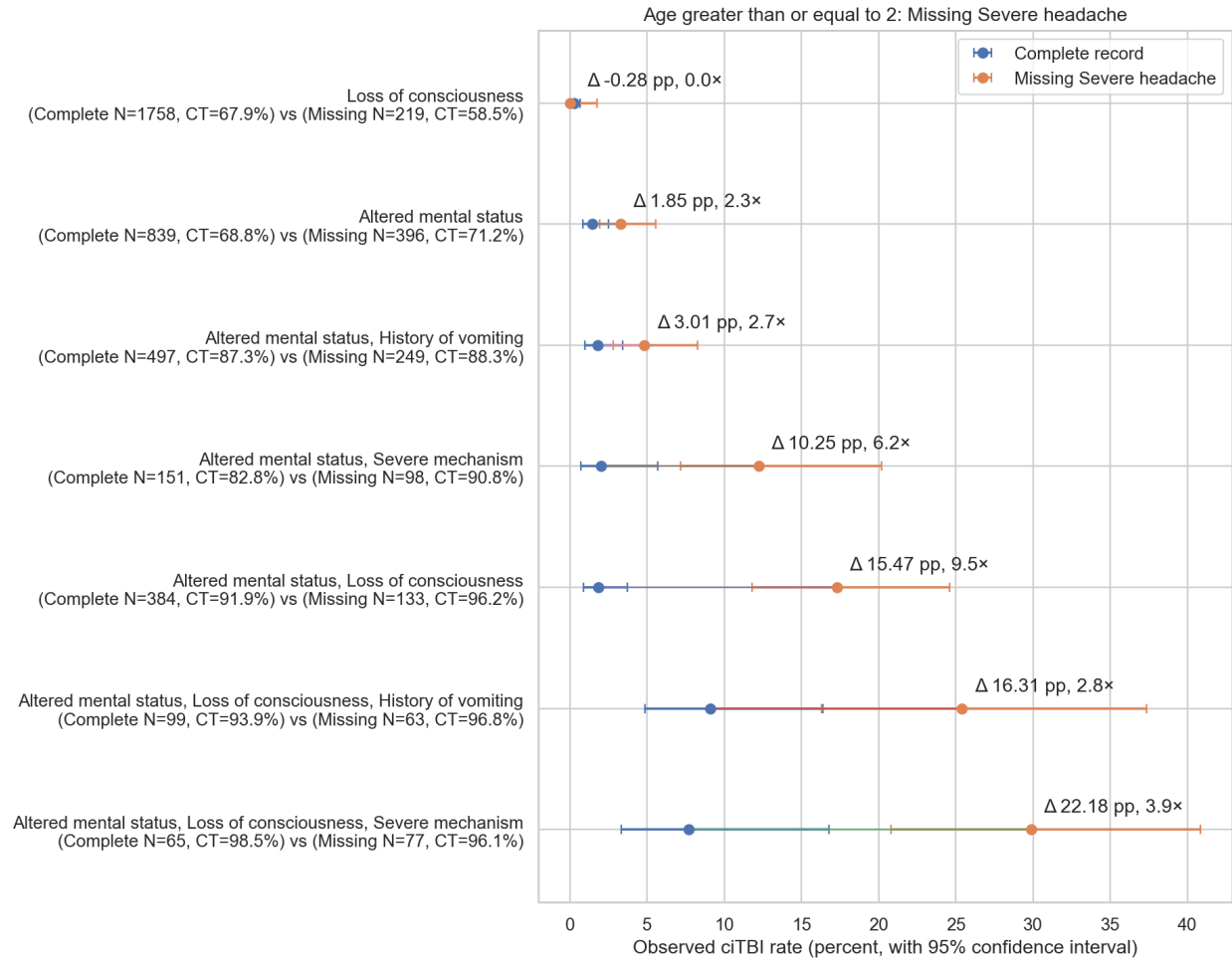
Several PECARN variables use a parent field plus detail fields. A common inconsistency occurs when the parent indicates a symptom is present, but every detail field is missing. This creates ambiguity. The symptom may be real with incomplete documentation, or the parent may reflect uncertainty rather than a confirmed finding. I tested whether this documentation ambiguity drives the main result from Finding 1.

The cleaning pipeline controls this with `enforce_detail_presence`. In the baseline setting, I keep the parent value even if all details are missing. In the stricter perturbation, if the parent is "Yes" but all detail fields are missing, I recode the parent to NA. I treat this as a reasonable alternative because it makes the dataset internally consistent and prevents "Yes" labels that have no supporting details from being treated as fully observed.

Table 7: Table 7. Audit of rows changed to NA under strict detail presence (`enforce_detail_presence=True`) perturbation.

	parent	# Rows changed to NA	Percentage
0	History of LOC	2,554	588.76%
1	Headache	358	82.53%
2	Hematoma	86	19.83%
3	Skull Fracture Palpable	55	12.68%
4	Vomiting	54	12.45%
5	Seizure	52	11.99%

The audit table shows this perturbation mainly affects one variable: history of loss of consciousness (LOC). About 2,554 rows (roughly 5–6% of the dataset) change from observed to NA for `loc_separate`, while other parent variables change far less. This matters because the derived PECARN predictor "LOC" depends on this parent field and its duration detail, so the strict rule moves many borderline LOC cases into an "unknown LOC" state.



- AMS + LOC: still a large positive shift (baseline Δ 17.81 pp vs perturbed Δ 15.47 pp).
- AMS + LOC + vomiting: still large (baseline Δ 19.82 pp vs perturbed Δ 16.31 pp).
- AMS + LOC + severe mechanism: still very large (baseline Δ 26.81 pp vs perturbed Δ 22.18 pp).

The main qualitative conclusion remains. In the age 2 or more plot, high-acuity profiles involving altered mental status keep the same ordering and show similar ciTBI separation under missing severe headache. For example, the largest “missing severe headache” shifts remain large even after the perturbation, though slightly smaller in magnitude.

What changes most concentrates in LOC-only profiles. In the age 2 or more missing-severe-headache plot under perturbation, the LOC-only comparison shrinks and flips sign relative to baseline.

- Baseline: “LOC” vs “LOC | missing severe headache” had Δ +1.21 pp.
- Perturbed: it becomes Δ -0.28 pp.

However, this does not contradict the main finding. It shows that subgroup estimates tied to LOC alone are sensitive to how incomplete LOC documentation is handled, which is expected because the perturbation almost exclusively targets LOC. Overall, this stability check supports two conclusions. First, the core Finding 1 result survives a defensible documentation perturbation: missingness remains associated with higher observed ciTBI in clinically concerning profiles, especially those involving altered mental status. Second, LOC-only profile estimates are less stable because the strict rule redefines who counts as “LOC present.”

Part II: Predictive Modeling

I fit three models that recommend CT (yes or no) using clinical information available at presentation, and I evaluate them against clinically important TBI (ciTBI). The main clinical cost is missing a ciTBI, so sensitivity and the number of false negatives are the primary metrics. I also report specificity and NPV, and I track CT recommendation rate as a proxy for CT utilization. Because PECARN uses different criteria by age, I fit and evaluate models separately for children under 2 and children 2 years and older. For the learned models, I use an 80/20 stratified train-test split, and I choose the decision threshold using cross-validated predictions on the training set only.

ciTBI is a binary outcome defined as any of the following: neurosurgery, intubation longer than 24 hours due to head trauma, death due to TBI, or hospitalization for at least two nights for a head injury with TBI on CT.

Model 1: PECARN clinical decision rule (Kuppermann et al.)

I implemented the published PECARN clinical decision rule as a baseline. The rule is a short checklist that recommends CT when any age-specific risk factor is present, which makes it easy to apply and interpret.

For under 2, the rule recommends CT if any of altered mental status, non-frontal scalp hematoma, loss of consciousness at least 5 seconds, severe mechanism, palpable skull fracture, or not acting normally is present. For ages 2 and older, it recommends CT if any of altered mental status, any loss of consciousness, vomiting, severe mechanism, signs of basilar skull fracture, or severe headache is present. In my implementation, missing symptom values are treated as absent so that every patient receives a decision.

Model 2: Logistic Regression

I fit a penalized logistic regression model to produce a continuous risk score and then convert it to a CT recommendation using a threshold. I impute missing values with 0 and add missingness indicators, which matches a “charting by exception” workflow and allows the model to learn when documentation gaps carry signal. I tune regularization using cross-validated ROC AUC on the training set, then select the operating threshold using out-of-fold training predictions.

In under-2 children, logistic regression recommends CT for 24.5% of patients and has specificity 0.763, but sensitivity is only 0.806. This operating point would reduce CT substantially, but it would miss too many ciTBI cases relative to PECARN. In the 2+ group, logistic regression achieves sensitivity 0.967 with CT rate 0.265 and specificity 0.749, suggesting it can match PECARN-level sensitivity in this split while recommending fewer CTs.

Model 3: Decision Tree

I fit a decision tree to capture nonlinear interactions and threshold behavior. I use the same imputation and missingness indicators as logistic regression. I tune tree depth and pruning using cross-validated ROC AUC on the training set and select the threshold using out-of-fold training predictions.

In under-2 children, the tree increases sensitivity relative to logistic regression (0.935 vs 0.806) but still falls short of PECARN (0.990). Its CT rate is 0.333, between logistic regression and PECARN. In the 2+ group,

the tree matches logistic regression sensitivity (0.967) but with lower specificity (0.573) and higher CT rate (0.438), close to PECARN utilization in this split. Feature importances show the tree relies most on altered mental status and other neurologic severity signals, with skull fracture indicators and injury severity also contributing.

Model Performance Summary

PECARN provides the most clinically aligned operating point across age groups, with very high sensitivity and moderate CT utilization. In under-2 children, neither learned model reaches PECARN sensitivity at the selected thresholds, which is the key limitation for clinical use. In children aged 2 and older, logistic regression achieves PECARN-like sensitivity with a lower CT rate in this split, while the decision tree achieves similar sensitivity but does not reduce CT use.

Table 8: Table 8. Performance metrics for PECARN CDR, logistic regression, and decision tree models by age group.

	Age Group	Model	Threshold	Sensitivity	Specificity	NPV	CT Rate
0	Two Plus	PECARN CDR		0.968	0.578	1.000	0.427
1	Two Plus	Logistic regression	0.173000	0.967	0.749	0.999	0.265
2	Two Plus	Decision tree	0.132000	0.967	0.573	0.999	0.438
3	Under 2	PECARN CDR		0.990	0.539	1.000	0.466
4	Under 2	Logistic regression	0.171000	0.806	0.763	0.996	0.245
5	Under 2	Decision tree	0.083000	0.935	0.675	0.999	0.333

Interpretability

PECARN is easy to read and use. It works as a checklist with a simple OR rule. A CT is recommended if any high-risk feature is present. Logistic regression acts like a weighted score. Each finding raises or lowers predicted risk. The decision threshold determines when the recommendation switches to CT. If we include missingness indicators, part of the signal may reflect documentation gaps rather than true clinical risk. In that case, a “high risk” driven by missing data should prompt a more complete assessment. A decision tree reads like a flowchart. We can trace the path and see exactly which variables led to the recommendation. However, trees are unstable. Small changes in preprocessing or missing data handling can alter splits and decision paths, so we should check stability before trusting a single fitted tree.

Stability check

I repeated the full modeling pipeline after applying a perturbation to the cleaning process by setting `enforce_detail_presence=True`. This changes how records behave when a “parent” symptom is recorded but required detail fields are missing. I compare baseline and perturbed results using sensitivity, false negatives, CT rate, and specificity. Logistic regression is stable. Its metrics change only slightly in both age groups, and under-2 sensitivity and false negatives are unchanged. The decision trees are less stable. In under-2 children, tree sensitivity drops from 0.935 to 0.871 and false negatives increase from 2 to 4, while CT rate is nearly unchanged. In the 2+ group, the tree shifts toward fewer CT recommendations, with CT rate dropping from 0.438 to 0.369, but this comes with a small sensitivity decrease. PECARN is stable for under 2 but not for 2+.

Under 2, its metrics are unchanged. For ages 2 and older, sensitivity drops from 0.968 to 0.914 while CT rate decreases. This indicates that the perturbation changed the effective definition of at least one rule input in the 2+ group, shifting the rule to recommend fewer CTs at the cost of more missed ciTBI.

Table 9: Table 9. Stability check results comparing baseline and perturbed models across sensitivity, CT rate, specificity, and false negatives by age group and model type.

	Age Group	Model	Sensitivity Δ	CT Rate Δ	Specificity Δ
0	Two Plus	Decision tree	-0.008	-0.068	0.069
1	Two Plus	Logistic regression	0.008	0.007	-0.007
2	Two Plus	PECARN CDR	-0.054	-0.031	0.031
3	Under 2	Decision tree	-0.065	0.002	-0.003
4	Under 2	Logistic regression	0.000	0.004	-0.004
5	Under 2	PECARN CDR	0.000	0.000	0.000

Discussion

The dataset includes 42,412 pediatric patients, so size was not the main challenge. The real issue was data quality. Many variables had substantial missingness, and the outcome, ciTBI, was very rare at only 0.89%. This lab navigates across all three realms of data science. The real-world realm is the physical pediatric emergency department, where children present with head trauma and clinicians make decisions. The data realm is the PECARN dataset itself. The algorithmic realm is the modeling part of the lab. There is definitively no one-to-one correspondence between this data and reality. Ultimately, without rigorous cleaning and critical visualization to evaluate these gaps, applying algorithms to this messy approximation would lead to untrustworthy, biased models that fail to capture true clinical reality.

Academic Honesty Statement

To Bin,

I pledge that all work submitted throughout this semester is my own and that every source used has been properly cited. Academic research honesty is necessary because it ensures the reproducibility of our results and maintains the collective trust required for scientific progress. Because without a commitment to truth, the data we analyze loses its meaning and its ethical foundation. I affirm that the analysis in this report is a transparent and honest reflection of my own understanding and effort.

Works Cited

Kuppermann, Nathan et al. "Identification of children at very low risk of clinically-important brain injuries after head trauma: a prospective cohort study." *Lancet* (London, England) vol. 374,9696 (2009): 1160-70. doi:10.1016/S0140-6736(09)61558-0

Yu, Bin, and Rebecca L. Barter. *Veridical Data Science: The Practice of Responsible Data Analysis and Decision Making*. MIT Press, 2023, vdsbook.com.

Use of AI Tools

I acknowledge the use of ChatGPT and GitHub Copilot to assist with code structure and organization. All generated suggestions were carefully reviewed, tested, and refined. I take full responsibility for the final implementation, modeling decisions, and interpretations presented in this report.